Investigation: AMERICANAS S.A. (CVM code 20990), fiscal year 2021, as it was on 2022-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Comércio (Atacado e Varejo) (origin: FCA); group: non-financial.
Trading codes (2021 FCA, version 3, filed on 2022-03-29): BTOW3 (Ações Ordinárias, Bolsa, since 2007-08-08).

Document used

2021 DFP, version 1, filed on 2022-02-24; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2022-02-24).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=112049

Possible manipulation (engine v1, study 1)

Beneish M-score (mscore)
- Value: -0.240 (index)
- Alert by the study 1 rule: yes
- Rule: M > -1.78 (pre-registration, section 4)
- Percentile in sector and year: 94 (n = 39)
- Percentile in group and year: 93 (n = 481)
- The company's own history: median -2.486; Tukey range (1.5 × IQR): from -5.212 to 0.647; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium
- Alternative explanations: In 2021, B2W (CVM code 20990) received the Lojas Americanas operations through a partial spin-off (shareholders' meeting of 2021-06-10); the 2020 comparative is B2W alone, and part of SGI comes from this change of perimeter (docs/en/results.md, Americanas section).

DSRI, days' sales in receivables index (DSRI)
- Value: 1.703 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 90 (n = 43)
- Percentile in group and year: 92 (n = 576)
- The company's own history: median 0.841; Tukey range (1.5 × IQR): from -0.602 to 2.942; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index (GMI)
- Value: 0.954 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 26 (n = 44)
- Percentile in group and year: 41 (n = 594)
- The company's own history: median 0.905; Tukey range (1.5 × IQR): from 0.677 to 1.188; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index (AQI)
- Value: 1.213 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 81 (n = 44)
- Percentile in group and year: 82 (n = 583)

The company's own history: median 0.884; Tukey range (1.5 × IQR): from 0.618 to 1.289; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGI, sales growth index (SGI)
Value: 2.242 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 44)
Percentile in group and year: 95 (n = 595)
The company's own history: median 1.032; Tukey range (1.5 × IQR): from 0.821 to 1.176; value above the range (index; 5 fiscal years)
Confidence in the data: medium
Alternative explanations: In 2021, B2W (CVM code 20990) received the Lojas Americanas operations through a partial spin-off (shareholders' meeting of 2021-06-10); the 2020 comparative is B2W alone, and part of SGI comes from this change of perimeter (docs/en/results.md, Americanas section).

DEPI, depreciation index (DEPI)
Value: 3.403 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 99 (n = 40)
Percentile in group and year: 99 (n = 500)
The company's own history: median 0.847; Tukey range (1.5 × IQR): from 0.743 to 1.012; value above the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index (SGAI)
Value: 1.014 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 60 (n = 44)
Percentile in group and year: 64 (n = 590)
The company's own history: median 1.078; Tukey range (1.5 × IQR): from 0.672 to 1.475; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets (TATA)
Value: 0.031 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 74 (n = 45)
Percentile in group and year: 70 (n = 617)
The company's own history: median -0.007; Tukey range (1.5 × IQR): from -0.078 to 0.051; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index (LVGI)
Value: 0.983 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 31 (n = 44)
Percentile in group and year: 43 (n = 603)
The company's own history: median 0.966; Tukey range (1.5 × IQR): from 0.771 to 1.122; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump (dpo)
Value: -6.8 days
Alert by the study 1 rule: no

Rule: DPO change in the year's top decile among non-financial companies (>= 90th percentile of computable values on the date: 28.9 days)
Percentile in sector and year: 42 (n = 44)
Percentile in group and year: 36 (n = 566)
The company's own history: median 4.4 days; Tukey range (1.5 × IQR): from -31.7 days to 54.2 days; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI (juro)
Value: 9.42 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 35 (n = 41)
Percentile in group and year: 47 (n = 557)
The company's own history: median 10.64 p.p.; Tukey range (1.5 × IQR): from -1.43 p.p. to 26.45 p.p.; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets (lucro_sem_caixa)
Value: 0.041 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 77 (n = 45)
Percentile in group and year: 73 (n = 614)
The company's own history: median -0.008; Tukey range (1.5 × IQR): from -0.086 to 0.056; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: 2.051 (index)
Components: X1 = 0.193; X2 = 0.027; X3 = 0.017; X4 = 0.556 (indices)
Percentile in sector and year: 46 (n = 45)
The company's own history: median 1.819; Tukey range (1.5 × IQR): from 1.379 to 2.342; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 5/9 (8 computable)
Signals: ROA = 1; CFO = 0; Delta ROA = 1; ACCRUAL = 0; Delta LEVER = 1; Delta LIQUID = 0; EQ_OFFER = —; Delta MARGIN = 1; Delta TURN = 1
- EQ_OFFER from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2016, v1: 2017-02-20
- DFP 2017, v1: 2018-03-07
- DFP 2018, v1: 2019-03-20
- DFP 2019, v1: 2020-02-20
- DFP 2019, v2: 2020-02-21
- DFP 2020, v1: 2021-03-04
- DFP 2020, v2: 2021-03-04
- FCA 2021, v1: 2021-05-31
- FCA 2021, v2: 2021-06-21
- DFP 2021, v1: 2022-02-24
- FCA 2021, v3: 2022-03-29

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Americanas S.A. e da Americanas S.A. e suas controladas em 31 de dezembro de 2021, o desempenho de suas operações e os seus respectivos fluxos de caixa, bem como o desempenho consolidado de suas operações e os seus fluxos de caixa consolidados para o exercício findo nessa data, de acordo com as práticas contábeis adotadas no Brasil e as normas internacionais de relatório financeiro (IFRS) emitidas pelo International Accounting Standards Board (IASB)."
Auditor in the 2019 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor in the 2020 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor in the 2021 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2011: Ernst & Young Auditores Independentes S.S. -> PricewaterhouseCoopers Auditores Independentes (transition: two firms in the 2011 FCA). The outgoing firm audited 2 consecutive fiscal years (2009 to 2010), by the start and end dates in the FCA.
- 2016: PricewaterhouseCoopers Auditores Independentes -> KPMG Auditores Independentes. The outgoing firm audited 5 consecutive fiscal years (2011 to 2015), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2019: KPMG Auditores Independentes -> PricewaterhouseCoopers Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2016 to 2018), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-

years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-09T19:18:01+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.